

Feature Extraction:-

A feature is a distinctive attribute or description of something we want to label or differentiate, more generally are going to be of value in differentiating between entire images or families of images.

Two principal aspects of image feature extraction $\begin{cases} \rightarrow \text{feature detection} \\ \rightarrow \text{feature description} \end{cases}$

The extraction process must encompass both.

A feature description is invariant w.r.t. a set of transformations if its value remains unchanged after the application of any transformation from the family.

It is covariant w.r.t. a set of transformations if applying to the entire any transformation from the set produces the same result in the description.

When n descriptors are used, feature vectors become n -dimensional and the space containing them is referred to as an n -dimensional feature space.

We group features into 3 principal categories —
① Boundary ② Region ③ whole image features.

Boundary

① Boundary following / Tracing

We assume that we are working with binary images where object background points are labeled 1 and 0 respectively.

② The images are padded with a border of 0's to eliminate the possibility of an object merging with the image border.

Moore Boundary Tracing Algorithm

① Find top leftmost foreground pixel, let b_0 be such pixel, c_0 be the west neighbour of b_0 , c_0 is always a background point.

② Let $b = b_0$ and $c = c_0$

③ Let the 8-neighbours of b , starting at c and proceeding in a clockwise direction be denoted by $(n_1, n_2, \dots, n_8)$. Find the first neighbour labeled 1 (foreground) and denote it by n_k .

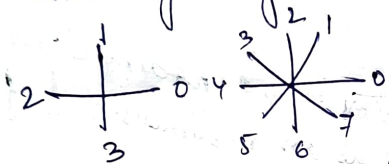
④ Let $b = n_k$, $c = n_{k-1}$.

- 5) Repeat step 3 and 4 until $b = b_0$. The sequence of b points found when the algorithm stops, is the set of ordered boundary points.

Chain Codes

Chain codes are used to represent a boundary by a connected sequence of straightline segments of specified length and direction.

Typically a chain code representation is based on 4 or 8 connectivity of the segments. The direction of each segment is coded by using a numbering scheme.



A boundary code formed as a sequence of such directional numbers is referred to as a Freeman chain code.

This level of detail generally is not used for two principal reasons —

- (i) The resulting chain would be quite long.
- (ii) Any small disturbances along the boundary due to noise / imperfect segmentation would cause changes in the code that may not be related to the principal shape features of the boundary.

Solution $\Rightarrow$ Instead of fine grid, divide the image into larger rectangular grid. Then as the boundary is traversed, a boundary point is assigned to a node of the coarser grid, depending on the proximity of the original boundary point to that node.

- (iii) Depending on starting point, code varies
 Solⁿ $\Rightarrow$ Considering the code is circular, apply cyclic shift and consider numerically lowest code.

(iv) Code dependent on object size.
Solⁿ $\rightarrow$ make the object boundary scale to some fixed size and then chain code.

(v) Code is rotation variant,
Solⁿ $\rightarrow$ Rotation invariant

Normalize by using the first difference of the chain code instead of code itself. This difference is obtained by counting the no. of direction changes that separate two adjacent elements of the code.

Skeleton

Skeletons can be computed from a boundary by filling the area enclosed by the boundary with foreground values and treating the result as a binary region. A skeleton is computed using the coordinates of points in the entire region, including its boundary.

The skeleton is the set of points in the region that are equidistant from the border of the region.

Two principal approaches $\rightarrow$ (i) By successively thinning the region while preserving end points and line connectivity (topology preserving thinning).

(ii) By computing the medial axis of the region via an efficient implementation of the medial axis transform (MAT).

Let R be a region and B is its boundary. For each point P in R , we find its closest neighbour in B . If P has more than one such neighbour, it is said to belong to the medial axis of R .

Computational heavy. Instead approach is to obtain the skeleton equivalently from the distance transform.

The distance transform of a region of foreground pixels in a background of zeros is the distance from every pixel to the nearest non zero valued pixel.

for each foreground pixel find distance to the nearest background pixels, which constitute the region boundary.
 Thus we compute the transform of the complement of the image.

Thinning:- (single pixel width generated)

Search Process $\rightarrow$ A pixel can be deleted if it is not a critical point or not an end point.

1	0	0
0	1	0
0	0	1

$\rightarrow$ end point (deletion leads to change the object length)

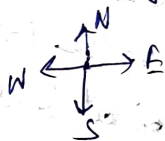
0	1	0
0	1	0
0	1	0

$\rightarrow$ Critical point (deletion leads to discontinued local neighbourhood)

To maintain the symmetry, all the non-critical and non-end points to be deleted simultaneously. [2 pixel width objects may vanish]

$\Downarrow$

Instead of this consider NESW
 (operate one side at a time)



Repeat

{ for each side $i \in \{N, E, S, W\}$

{ for each pixel in the side

{ search and delete non critical and non-end points

{ }

{ till no change.

Thinning is the method for extracting single pixel thick connected skeleton that preserves basic shape information, such skeletons faithfully represent shape of objects that are mainly composed of ribbon-like structures.

Texture

An important approach to region description is to quantify its texture content. With no formal definition of texture exists, intuitively, this descriptor provides measures of properties such as smoothness, coarseness, regularity etc.

It is captured as intensity variation in object/image. This variation appears as repetition of basic pattern.

Basic patterns of large size $\rightarrow$ coarser texture
" " " " smaller size $\rightarrow$ finer texture.

Also, Repetition may or may not be random

GLCM (Gray-level co-occurrence matrix)

Distribution of joint occurrence of pairs of intensity values at a specified distance & orientation.

Since basic patterns are formed by certain arrangements of gray levels, co-occurrence of gray levels at predefined relative positions may be an indication of presence and periodicity of basic pattern.

Let $\mathcal{Q}$ be an operator that defines the position of two pixels relative to each other and consider an image f with L possible intensity levels. Let $G_{\mathcal{Q}}$ be a matrix whose element g_{ij} is the no. of times that pixel pairs with intensities z_i and z_j occur in image f in the position specified by $\mathcal{Q}$ where $1 \leq i, j \leq L$.

A matrix is formed in this manner is referred to as a GLCM.

The total no. of pixel pairs (n) that satisfy $\mathcal{Q}$ is equal to the sum of the elements of $G_{\mathcal{Q}}$, then the quantity

$P_{ij} = \frac{g_{ij}}{n}$ is an estimate of the probability

that a pair of points satisfying $\mathcal{Q}$ will have values (z_i, z_j) .
 $K = \text{row and column dimension of square matrix } G_{\mathcal{Q}}$.

$$\sum_{i=1}^K \sum_{j=1}^K P_{ij} = 1$$

As G_c depends on G , the presence of intensity texture patterns can be detected by choosing an appropriate position operator and analyzing the elements of G_c .

$$m_r = \sum_{i=1}^K i \sum_{j=1}^K p_{ij}$$

$$m_c = \sum_{j=1}^K j \sum_{i=1}^K p_{ij}$$

Maximum probability $\rightarrow \max_{i,j} (p_{ij})$

measures the strongest response of G_c

Correlation $\rightarrow \sum_{i=1}^K \sum_{j=1}^K \frac{(i-m_r)(j-m_c)p_{ij}}{G_r G_c}$

A measure of how correlated a pixel is to its neighbours over the entire image.

$$G_r \neq 0 \quad G_c \neq 0$$

Contrast $\rightarrow \sum_{i=1}^K \sum_{j=1}^K (i-j)^2 p_{ij}$

A measure of intensity contrast between a pixel and its neighbours over the entire image.

Energy $\rightarrow \sum_{i=1}^K \sum_{j=1}^K p_{ij}^2$

$$G_r^2 = \sum_{i=1}^K (i-m_r)^2 \sum_{j=1}^K p_{ij}$$

Homogeneity $\rightarrow \sum_{i=1}^K \sum_{j=1}^K \frac{p_{ij}}{1+|i-j|}$

$$G_c^2 = \sum_{j=1}^K (j-m_c)^2 \sum_{i=1}^K p_{ij}$$

Entropy $\rightarrow - \sum_{i=1}^K \sum_{j=1}^K p_{ij} \log_2 p_{ij}$

Measures the randomness of the elements of G_c

Moment $\rightarrow \sum_{i=1}^K \sum_{j=1}^K i^m j^n p_{ij}$

Local Binary Pattern (LBP)

For each pixel, generate a pattern by looking into its 8-neighbourhood at intensity at (i,c) by μ if $f(i,c) \geq \mu \rightarrow (i,j)$ th nbd of (i,c) else $f'(i,j) = 0$

1	0	1
1	(i,c)	1
0	0	1

(0-255) $\rightarrow$ 256 possible patterns

$\rightarrow$ Prepare a 256-dimension histogram of LBPs of the pixels
 $\rightarrow$ Normalize